

3.5.2 Selection Statements: if-elif-else and Nested if

if-elif-else میں کئی شرطیں اوپر سے نیچے ترتیب سے دیکھی جاتی ہیں، اور جو شرط سب سے پہلے درست ہو اسی کا بلاک چلتا...

CS XI

Chapter 3: Programming Fundamentals

Lecture 16 of 24

★ Golden

3.5.2 Selection Statements: if-elif-else and Nested if ★ Golden

LEARN IT

3. The if-elif-else statement handles **many conditions**. (`elif` is short for "else if".) Python checks the conditions **from top to bottom**. As soon as it finds a True condition, it runs that block and **ignores all the rest**. If none is True, the `else` block runs.

4. Nested if is an `if` inside another `if`. It is used when a decision needs to pass **several levels of checks**: the inner check happens only if the outer check passed.

WORKED EXAMPLE

Grading with if-elif-else.

```
marks = int(input("Enter marks: "))
if marks >= 80:
    print("A-1 Grade")
elif marks >= 70:
    print("A Grade")
elif marks >= 60:
    print("B Grade")
else:
    print("Fail or below B")
```

Sample run (user types 75):

OUTPUT

```
Enter marks: 75
A Grade
```

Trace for different marks:

marks	>= 80?	>= 70?	>= 60?	Output
92	True	(not checked)	(not checked)	A-1 Grade
75	False	True	(not checked)	A Grade
60	False	False	True	B Grade
41	False	False	False	Fail or below B

Loan approval with nested if.

```
age = 20
salary = 35000
if age ≥ 18:
    # passed first check
    if salary ≥ 30000:
        # passed second check
        print("Loan APPROVED!")
    else:
        print("Salary too low for loan.")
else:
    print("Too young for loan.")
```

OUTPUT

Loan APPROVED!

age	salary	Outer check (age ≥ 18)	Inner check (salary ≥ 30000)	Output
20	35000	True	True	Loan APPROVED!
20	25000	True	False	Salary too low for loan.
16	35000	False	(never checked)	Too young for loan.

When you only need a yes/no answer, two levels can often be joined with `and` — but then you lose the separate messages:

```
age = 20
salary = 25000
if age ≥ 18 and salary ≥ 30000:
    print("Loan APPROVED!")
else:
    print("Loan not approved.")
```

OUTPUT

Loan not approved.

REMEMBER IT

In an `if-elif-else` ladder, put the **strictest condition first** (highest marks first). Think of a ladder: you step off at the first rung that fits.

COMMON MISTAKE

Wrong order gives wrong answers. Here 85 marks gets "B Grade", because `85 ≥ 60` is True first and the other conditions are ignored:

```
marks = 85
if marks ≥ 60:
    print("B Grade")
elif marks ≥ 70:
    print("A Grade")
elif marks ≥ 80:
    print("A-1 Grade")
```

OUTPUT

B Grade

This is a **logical error**: no crash, just a wrong result. Fix it by checking `≥ 80` first.

if-elif-else میں کئی شرطیں اوپر سے نیچے ترتیب سے دیکھی جاتی ہیں، اور جو شرط سب سے پہلے درست ہو اسی کا بلاک چلتا ہے، باقی سب چھوڑ دی جاتی ہیں۔ نیسٹڈ ایف (Nested if) میں ایک if دوسرے if کے اندر ہوتا ہے، اور اندر والی شرط تب ہی دیکھی جاتی ہے جب باہر والی شرط درست ہو۔

EXAM TIP

Grade-calculator programs (if-elif-else) and "largest of three numbers" are very common long questions. Check conditions in the right order, indent nested blocks carefully (8 spaces for the inner block), and give a sample run. Tracing tables like the ones above earn full marks in "predict the output" questions.

CHECK YOURSELF

1. Predict the output:

```
temp = 30
if temp > 35:
    print("Hot")
elif temp > 25:
    print("Warm")
elif temp > 15:
    print("Pleasant")
else:
    print("Cold")
```

2. Predict the output:

```
x = 15
if x > 10:
    if x > 20:
        print("A")
    else:
        print("B")
else:
    print("C")
```

Answers: 1. Warm (30 > 35 is False; 30 > 25 is True, so the rest are skipped even though 30 > 15 is also True). 2. B (outer check 15 > 10 passes; inner check 15 > 20 fails).

Next lecture: 3.5.3 Repetition Statements: the for Loop and range()